GHSCI output data dictionary

All potential indicators and output variables

Generated 16 August 2026 using the Global Healthy and Sustainable City Indicators (GHSCI) tool (https://healthysustainablecities.github.io/).

This reference describes the output variables that can be produced through the combined GHSCI workflow, including optional GTFS public transport, Earth Engine (large public green space and urban heat vulnerability), cycling accessibility and longitudinal series analyses, across the scales of calculation (sample point, grid, city, custom aggregation areas, and longitudinal series panels). Permutable elements of variable names are represented by parameter placeholders — [destination], [x] (distance threshold) and [network] (cycling route stress measure) — described under Naming conventions and parameters, rather than exhaustively listing every permutation. Regional customisation (e.g. custom destinations, distance thresholds or aggregation areas) yields analogous variables following the same naming patterns; the data dictionary generated alongside each processed study region enumerates the variables actually calculated for that region.

Study region information

Indicator	Variable	Units	Statistic	Scale
Continent	Continent		category	city
Country	Country		category	city
2-letter country code	ISO 3166-1 alpha-2		category	city
Study region	study_region		category	city, grid

Derived study region statistics

Indicator	Variable	Units	Statistic	Scale
Area (km^2; accounting for urban restrictions, if applied)	Area (sqkm)	km^2	sum	city
Population estimate, as per configured population data source	Population estimate	persons	sum	city
Population per km^2	Population per sqkm	persons per km^2	rate	city
Intersection count (following consolidation based on intersection tolerance parameter in region configuration)	Intersections	count	count	city
Intersections per km^2	Intersections per sqkm	count per km^2	rate	city
Area (km^2)	area_sqkm	km^2	sum	grid, custom areas
Population estimate, as per configured population data source	pop_est	persons	sum	grid, custom areas
Population per km^2	pop_per_sqkm	persons per km^2	rate	grid, custom areas
Intersection count (following consolidation based on intersection tolerance parameter in region configuration)	intersection_count	count	count	grid, custom areas
Intersections per km^2	intersections_per_sqkm	count per km^2	rate	grid, custom areas

Analytical statistics

Indicator	Variable	Units	Statistic	Scale
Sample points used in this analysis (generated along pedestrian network for populated grid areas)	urban_sample_point_count	count	count	city, grid
Population grid cell identifier	grid_id		category	grid, sample point
Sample point identifier	point_id		category	sample point
Count of population grid cells associated with this area	grid_count	count	count	custom areas

Naming conventions and parameters

Many indicator variable names are assembled from a fixed stem (identifying the measure) and one or more of the permutable elements below, shown here as placeholders. For example, pop_pct_access_[x]m_[destination]_score and sp_cycle_[network]access_[destination]_[x]m. Each placeholder, its possible values and defaults are described below; regionally configured values follow the same naming patterns.

[destination]

The destination whose access is measured. The default destinations are listed below. Walking and cycling access draw on the same destination definitions, declared once in a region's "accessibility" configuration; each pairs a strict and a lenient variant of each core category (food, public open space, public transport) so that composite "all categories reachable" indicators can be derived per variant.

Destination	Variable stub	Category
Fresh food market / supermarket	fresh_food_market	Food (strict)
Fresh food, pooled (market, supermarket or convenience store)	fresh_food_pooled	Food (lenient)
Convenience store	convenience	Food
Large public open space (≥ 1.5 ha)	public_open_space_large	Open space (strict)
Any public open space	public_open_space_any	Open space (lenient)
Large public green space (≥ 1 ha; OpenStreetMap and Earth Engine)	large_public_green_space	Open space
Frequent public transport (≤ 20 min daytime service; GTFS)	pt_frequent	Transport (strict, cycling)
Public transport, ≤ 20 min daytime service (GTFS)	pt_gtfs_freq_20	Transport (walking)
Public transport, ≤ 30 min daytime service (GTFS)	pt_gtfs_freq_30	Transport (walking)
Any public transport stop (GTFS)	pt_gtfs_any	Transport
Any public transport stop (OpenStreetMap or custom)	pt_osm_any	Transport
Any public transport stop (best of GTFS and OpenStreetMap/custom)	pt_any	Transport (lenient)
All categories reachable, strict variants	all_strict	Combined
All categories reachable, lenient variants	all_lenient	Combined
Local activity centre (everyday destinations co-located)	activity_centre_local	Derived
Complete activity centre (high-amenity destinations co-located)	activity_centre_complete	Derived

Regions can define their own destinations (custom OpenStreetMap tags or supplied datasets), pool categories, and configure additional combined-access sets and activity-centre definitions (a network location whose walkable catchment, default 400 m, contains at least one destination of every required category). These follow the same naming patterns, e.g. all_<set>_<variant> for a named combined-access set and activity_centre_<definition>_<tier> for a named activity centre. Combined-access composites have access indicators only (no nearest-distance variables).

[x]

Distance threshold (m). Distances can be customised to locally policy-relevant values. The core walking access indicators use the configured neighbourhood accessibility threshold (default 500 m). Where a region configures the optional pedestrian accessibility analysis, walking access is additionally evaluated at each configured pedestrian distance, and cycling access at each configured cycling distance (defaults: 500 m, 1000 m, 2000 m and 5000 m), so an access variable exists for each distance.

[network]

Cycling route stress measure, appearing as an optional prefix on cycling variable names: "lts1_" for routes composed entirely of lowest-stress Level of Traffic Stress (LTS) 1 links (suitable for all ages and abilities); "safe_" for fully low-stress routes on LTS 1–2 links (the default headline measure); or no prefix for danger-weighted routing over the full cycling network (higher-stress links usable at a proportionate distance penalty). Which measures are computed is configurable (cycling_indicators contrasts and measures settings).

Indicator estimates

Access (walking)

Indicator	Variable	Units	Statistic	Scale
Score (0-1) for walking access within the configured accessibility threshold distance (default 500 m) to the destination	sp_access_[destination]_score	score 0-1	value	sample point
Distance (m) along the pedestrian network to the nearest destination	sp_nearest_node_[destination]	metres	value	sample point
Percentage of population with walking access within [x] m of the destination	pct_access_[x]m_[destination]_ score	percent	percenta ge	grid, custom areas
Percentage of population with walking access within [x] m of the destination (population weighted)	pop_pct_access_[x]m_ [destination]_score	percent	percenta ge	city, custom areas
Score (0/1): the destination is reachable within [x] m along the pedestrian network	sp_walk_access_[destination]_ [x]m	score 0-1	value	sample point
Distance (m) along the pedestrian network to the nearest destination	sp_walk_nearest_node_ [destination]	metres	value	sample point
Percentage of population with walking access within [x] m to the destination	pct_access_walk_[destination]_ [x]m	percent	percenta ge	grid, custom areas
Percentage of population with walking access within [x] m to the destination (population weighted)	pop_pct_access_walk_ [destination]_[x]m	percent	percenta ge	city, custom areas
Average distance (m) along the pedestrian network to the nearest destination	avg_walk_dist_[destination]	metres	mean	grid, custom areas
Average distance (m) along the pedestrian network to the nearest destination (population weighted)	pop_avg_walk_dist_[destination]	metres	mean	city, custom areas
Score (0/1): the nearest destination is further than [x] m along the pedestrian network	sp_walk_beyond_[destination]_ [x]m	score 0-1	value	sample point
Percentage of population living further than [x] m along the pedestrian network from the nearest destination	pct_beyond_walk_[destination]_ [x]m	percent	percenta ge	grid, custom areas
Percentage of population living further than [x] m along the pedestrian network from the nearest destination (population weighted)	pop_pct_beyond_walk_ [destination]_[x]m	percent	percenta ge	city, custom areas

Walkability

Indicator	Variable	Units	Statistic	Scale
Average population density of the local walkable neighbourhood (population per km², based on the population grid cells intersecting the sample point neighbourhood)	sp_local_nh_avg_pop_density	persons per km²	value	sample point
Average street intersection density of the local walkable neighbourhood (intersections per km²)	sp_local_nh_avg_intersection_ density	count per km²	value	sample point
Daily living score (/3): sum of access scores for fresh food, convenience and public transport destinations	sp_daily_living_score	score 0-3	value	sample point
Walkability index: sum of z-scores of the daily living score, neighbourhood population density and neighbourhood intersection density	sp_walkability_index	index (sum of z-scores)	value	sample point
Average walkable neighbourhood population density (population per km²) of sample points in this area	local_nh_population_density	persons per km²	mean	grid, city, custom areas

Indicator	Variable	Units	Statistic	Scale
Average walkable neighbourhood intersection density (intersections per km²) of sample points in this area	local_nh_intersection_density	count per km²	mean	grid, city, custom areas
Average daily living score (/3) of sample points in this area	local_daily_living	score 0-3	mean	grid, city, custom areas
Average walkability index of sample points in this area	local_walkability	index (sum of z-scores)	mean	grid, city, custom areas
Average walkable neighbourhood population density (population per km²; population weighted)	pop_nh_pop_density	persons per km²	mean	city, custom areas
Average walkable neighbourhood intersection density (intersections per km²; population weighted)	pop_nh_intersection_density	count per km²	mean	city, custom areas
Average daily living score (/3; population weighted)	pop_daily_living	score 0-3	mean	city, custom areas
Average walkability index (population weighted)	pop_walkability	index (sum of z-scores)	mean	city, custom areas

Cycling accessibility

Indicator	Variable	Units	Statistic	Scale
Score (0/1): the destination is reachable within [x] m by cycling, routed using the [network] measure	sp_cycle_[network]access_ [destination]_[x]m	score 0-1	value	sample point
Distance (m) by cycling to the nearest destination, routed using the [network] measure	sp_cycle_[network]nearest_ node_[destination]	metres	value	sample point
Percentage of population with cycling access within [x] m to the destination, routed using the [network] measure	pct_access_cycle_ [network][destination]_[x]m	percent	percenta ge	grid, custom areas
Percentage of population with cycling access within [x] m to the destination, routed using the [network] measure (population weighted)	pop_pct_access_cycle_ [network][destination]_[x]m	percent	percenta ge	city, custom areas
Average distance (m) by cycling to the nearest destination, routed using the [network] measure	avg_cycle_dist_ [network][destination]	metres	mean	grid, custom areas
Average distance (m) by cycling to the nearest destination, routed using the [network] measure (population weighted)	pop_avg_cycle_dist_ [network][destination]	metres	mean	city, custom areas

Urban heat vulnerability

Derived from Earth Engine raster analyses on a 1 km grid by default, following Turner et al. (2025), "Development and validation of the Global Urban Heat Vulnerability Index (GUHVI)", Urban Climate 64: 102716. Inputs are aligned so that higher values indicate greater heat vulnerability. Each measure below is sampled at sample points (sp_ prefix), averaged for each grid cell or custom area (no prefix), and population weighted for the city and custom areas (pop_ prefix). Sub-indicators are listed beneath the composite index they contribute to.

Indicator	Variable	Units	Statistic	Scale
Heat Exposure Index (HEI): the heat exposure sub-index, derived from land surface temperature	urban_heat_exposure_index	index 0-1	mean	sample point, grid, city, custom areas
Land surface temperature (°C): the temperature of the Earth’s surface, indicating the lived experience of heat (Landsat 8 surface reflectance, hottest third of the year)	urban_heat_land_surface_temp_ c	degrees Celsius	mean	sample point, grid, city, custom areas

Indicator	Variable	Units	Statistic	Scale
Heat Sensitivity Index (HSI): the equally weighted heat sensitivity sub-index of land surface albedo, NDVI, NDBI, local climate zone, population density and vulnerable population	urban_heat_sensitivity_index	index 0-1	mean	sample point, grid, city, custom areas
Land surface albedo: solar reflectance, where higher reflectance implies less surface heat retention (inverted, so that higher values indicate greater heat vulnerability)	urban_heat_land_surface_albedo	proportion 0-1	mean	sample point, grid, city, custom areas
Normalised Difference Vegetation Index (NDVI): a spectral index of vegetation health and greenness (inverted, so that higher values indicate greater heat vulnerability)	urban_heat_ndvi	index -1 to 1	mean	sample point, grid, city, custom areas
Normalised Difference Built-up Index (NDBI): a spectral index of impervious surfaces such as concrete and asphalt	urban_heat_ndbi	index -1 to 1	mean	sample point, grid, city, custom areas
Local Climate Zone: a standardised classification of land cover typologies for urban heat island studies, ordered from least heat retaining (water) to most heat retaining (bare rock or paved)	urban_heat_local_climate_zone_ordered	ordered class	category	sample point, grid, city, custom areas
Population density (persons per km²), from the GHS-POP global human settlement population grid	urban_heat_population_density_per_sqkm	persons per km²	mean	sample point, grid, city, custom areas
Vulnerable population (%): those most at risk from heat on the basis of age, being the combined proportion of the population aged 0–4 and 65+ years	urban_heat_vulnerable_pop_pct	percent	percentage	sample point, grid, city, custom areas
Adaptive Capability Index (ACI): the equally weighted adaptive capability sub-index of child dependency ratio, subnational Human Development Index and infant mortality rate; higher values indicate lower adaptive capability	urban_heat_adaptive_capability_index	index 0-1	mean	sample point, grid, city, custom areas
Child dependency ratio: children (aged 0–14 years) relative to the working age population (15–64 years), indicating economic reliance	urban_heat_child_dependency_ratio	ratio	mean	sample point, grid, city, custom areas
Subnational Human Development Index: human well-being assessed through education, health and standard of living	urban_heat_subnational_hdi	index 0-1	mean	sample point, grid, city, custom areas
Infant mortality rate: deaths of children under 1 year of age per 1000 live births, a key indicator of population health	urban_heat_infant_mortality_rate	deaths per 1,000 live births	mean	sample point, grid, city, custom areas
Global Urban Heat Vulnerability Index (GUHVI): the equally weighted composite of the Heat Exposure, Heat Sensitivity and Adaptive Capability indices, (HEI + HSI + ACI) / 3	urban_heat_guhvi	index 0-1	mean	sample point, grid, city, custom areas
Global Urban Heat Vulnerability Index (GUHVI) class, from 1 (least vulnerable) to 5 (most vulnerable)	urban_heat_guhvi_class	class 1-5	category	sample point, grid, city, custom areas
Percentage of population in Global Urban Heat Vulnerability Index (GUHVI) class 5, the most vulnerable class	pct_urban_heat_guhvi_class_5_most_vulnerable	percent	percentage	grid, city, custom areas

Longitudinal series outputs

Work in progress (July 2026)

Indicator	Variable	Units	Statistic	Scale
Timepoint label of the observation (e.g. year)	timepoint			longitudinal series
Output variable name of the indicator observed (see the indicator entries of this data dictionary)	indicator			longitudinal series
Value of the indicator at this timepoint for this unit	value			longitudinal series
Custom area identifier (area panels)	area_id			longitudinal series
Unit identifier (generic panels)	unit_id			longitudinal series
Baseline timepoint label of the comparison pair	t0			longitudinal series
Follow-up timepoint label of the comparison pair	t1			longitudinal series
Indicator value at the baseline timepoint	value_t0			longitudinal series
Indicator value at the follow-up timepoint	value_t1			longitudinal series
Change metric: 'pp_change' (percentage point change; used for bounded percentage indicators), 'diff' (absolute difference) or 'pct_change' (relative change, %, masked where the baseline is zero)	metric			longitudinal series
Change in the indicator between t0 and t1, as per the change metric	change			longitudinal series
Whether a policy review checklist was configured and scored for this timepoint	assessed			longitudinal series
Policy review checklist version	checklist_version			longitudinal series
Count of policies identified as present in the policy review	presence_numerator			longitudinal series
Number of policies assessed for presence in the policy review	presence_denominator			longitudinal series
Policy presence score (%)	presence_pct			longitudinal series
Sum of policy quality scores in the policy review	quality_numerator			longitudinal series
Maximum possible sum of policy quality scores in the policy review	quality_denominator			longitudinal series
Policy quality score (%)	quality_pct			longitudinal series